



levetivae® 500mg

film-coated tablets

Name and strength of active substance
Levetiracetam 500 mg

Product Description

levetivae 500 mg film-coated tablets are yellow, oblong, biconvex tablets.

Pharmacodynamics

Pharmacotherapeutic group: anti-epileptics, other anti-epileptics, ATC code: N03AX14.

The active substance, levetiracetam, is a pyrrolidone derivative (S-enantiomer of acetamide α-ethyl-2-oxo-1-pyrrolidine), chemically unrelated to any other existing anti-epileptic active substances.

Mechanism of Action

The mechanism of action of levetiracetam still remains to be fully elucidated. In vitro and in vivo experiments suggest that levetiracetam does not alter basic cell characteristics and normal neurotransmission. In vitro studies show that levetiracetam affects intraneuronal Ca²⁺ levels by partial inhibition of N-type Ca²⁺ currents and by reducing the release of Ca²⁺ from intraneuronal stores. In addition it partially reverses the reductions in GABA- and glycine-gated currents induced by zinc and β-carbolines. Furthermore, levetiracetam has been shown in in vitro studies to bind to a specific site in rodent brain tissue. This binding site is the synaptic vesicle protein 2A, believed to be involved in vesicle fusion and neurotransmitter exocytosis. Levetiracetam and related analogs show a rank order of affinity for binding to the synaptic vesicle protein 2A which correlates with the potency of their anti-seizure protection in the mouse audiogenic model of epilepsy. This finding suggests that the interaction between levetiracetam and the synaptic vesicle protein 2A seems to contribute to the antiepileptic mechanism of action of the medicinal product.

Pharmacodynamic effects

Levetiracetam induces seizure protection in a broad range of animal models of partial and primary generalised seizures without having a pro-convulsant effect. The primary metabolite is inactive. In man, an activity in both partial and generalised epilepsy conditions (epileptiform discharge/ photoparoxysmal response) has confirmed the broad spectrum pharmacological profile of levetiracetam.

Pharmacokinetics

Levetiracetam is a highly soluble and permeable compound. The pharmacokinetic profile is linear with low intra-and inter-individual variability. There is no modification of the clearance after repeated administration. There is no evidence of significant variability of gender, race or circadian. The pharmacokinetic profile in healthy volunteers and patients with epilepsy is comparable. Because absorption is complete and linear, plasma levels can be predicted, expressed as mg/kg of the body weight after oral administration of levetiracetam. Therefore, there is no need to monitor plasma levels of levetiracetam. In adults and children a significant correlation between the concentrations in saliva and plasma has been shown (ratio of saliva/plasma concentrations ranged from 1 to 1.7 for oral tablet formulation and after 4 hours post-dose for oral solution formulation).

Adults and adolescents

Absorption

Levetiracetam is rapidly absorbed after oral administration. The absolute oral bioavailability is close to 100%. The plasma concentration (C_{max}) peak is achieved 1.3 hours after administration. Steady plasma levels are achieved after two days with the twice daily administration schedule. Normal values of peak plasma concentrations (C_{max}) after a single dose of 1000 mg and repeated dose of 1000 mg twice daily are 31 and 43 µg/ml respectively. The extent of absorption is dose-independent and is not altered by food.

Distribution

There is no data for tissue distribution in humans. Neither levetiracetam nor its primary metabolite are significantly bound to plasma proteins (<10%). The volume of levetiracetam distribution is approximately 0.5 to 0.7 l/kg, close to the total body water volume.

Biotransformation

Levetiracetam is not extensively metabolised in humans. The main metabolic pathway (24% of dose) is an enzymatic hydrolysis of the acetamide group. The formation of the primary metabolite, ucb L057, is not supported by liver cytochrome P450 isoforms. The hydrolysis of the acetamide group was measurable in a number of tissues, including blood cells. The ucb L057 metabolite is pharmacologically inactive. Two minor metabolites were also identified. One was obtained by hydroxylation of the pyrrolidone ring (1.6% of the dose) and the other by the opening of the pyrrolidone ring (0.9% of the dose). Other unidentified compounds represented only 0.6% of the dose. No enantiomeric interconversion was evidenced in vivo for levetiracetam or its primary metabolite.

In vitro studies have shown that levetiracetam and its primary metabolite do not inhibit the major human liver cytochrome P450 (CYP3A4, 2A6, 2C9, 2C19, 2D6, 2E1 and 1A2), glucuronyl transferase (UGT1A1 and UGT1A6) and epoxide hydroxylase activity. In addition, levetiracetam does not affect the ***in vitro*** glucuronidation of valproic acid.

In human hepatocyte cultures, levetiracetam had little or no effect on CYP1A1/2, SULT1E1 or UGT1A1. Levetiracetam caused mild induction of CYP2B6 and CYP3A4. Data from in vitro and in vivo interactions with oral contraceptives, digoxin and warfarin indicate that there is not expected to significantly enzyme induction ***in vivo***. It is therefore highly unlikely for levetiracetam to interact with other substances, or vice versa.

Elimination

The plasma average life in adults was 7±1 hours and did not vary with dose, administration pathway or repeated administration. The average total body clearance was 0.96 ml/mm/kg. The major pathway of excretion was through the urinary tract, accounting for an average 95% of the dose (about 93% of the dose was excreted within the first 48 hours). Excretion via faeces accounted for only 0.3% of the dose. The cumulative urinary excretion of levetiracetam and its primary metabolite during the first 48 hours reached, respectively, 66% and 24% of the dose. The renal clearance of levetiracetam and ucb L057 is 0.6 and 4.2 ml/mm/kg respectively, indicating that levetiracetam is excreted through glomerular filtration with the subsequent tubular reabsorption and that the primary metabolite is also excreted by active tubular secretion in addition to glomerular filtration. Levetiracetam elimination is correlated to creatinine clearance.

Elderly patients

In the elderly, the average life is increased by about 40% (10-11 hours). This is related to the decrease in renal function in this population.

Kidney failure

The apparent body clearance is correlated with creatinine clearance for both levetiracetam and of its primary metabolite. Thus, in patients with moderate or severe kidney failure adjusting the maintenance daily dose of Levetiracetam based on creatinine clearance is recommended. In adult patients with anuric end-stage renal disease the average life was approximately 25 and 3.1 hours during interdialytic and intradialytic periods, respectively. The eliminated fraction of levetiracetam during a normal 4-hour dialysis was 51%.

Liver failure

In subjects with mild or moderate liver failure, there was no relevant clearance modification of the levetiracetam. In most subjects with severe liver failure, the clearance of levetiracetam was reduced by over 50% due to a concomitant kidney failure.

Paediatric group population

Children 4 to 12 years)

After administration of a single oral dose (20 mg/kg) to epileptic children (6-12 years), the average life of levetiracetam was 6.0 hours. The adjusted apparent body weight clearance was about 30% higher than in epileptic adults.

Following administration of multiple oral doses (20-60 mg/kg/day) to epileptic children (4 to 12 years), levetiracetam was rapidly absorbed. Peak plasma concentration was observed between 0.5 and 1.0 hours after administration. Dose linear and proportional increments for peak plasma concentrations and area under the curve increases were observed. Elimination average life was approximately 5 hours. The apparent body clearance was 1.1 ml/min/kg.

Breast-feeding infants and children (1 month to 4 years)

Following administration of single doses (20 mg/kg) of oral solution 100 mg/ml to epileptic children (1 month to 4 years), levetiracetam was rapidly absorbed and peak plasma concentrations were observed approximately 1 hour after administration. The pharmacokinetic results indicated that the average life was shorter (5.3 h) than for adults (7.2

hours) and apparent clearance was faster (1.5 ml/min/kg) than for adults (0.96 ml/min/kg). In the population pharmacokinetic analysis conducted in patients from 1 month to 16 years of age, body weight was significantly correlated to apparent clearance (clearance increased with increasing body weight) and the apparent volume of distribution. Age also had an influence on both parameters. This effect was pronounced for younger breast-feeding infants, and subsided as age increased, until it became negligible around 4 years old. In both population pharmacokinetic analyses, there was about a 20% increase of apparent clearance of levetiracetam when it was co-administered with an enzyme-inducing antiepileptic drugs.

Indication

levetivae is indicated as monotherapy in the treatment of partial onset seizures with or without secondary generalization in patients over 16 years with newly diagnosed epilepsy.

levetivae is indicated as an adjunctive treatment

- In the treatment of partial onset seizures with or without secondary generalization in adults, children from 4 years of age with epilepsy.
- In the treatment of myoclonic seizures in adults and children from 12 years with Juvenile Myoclonic Epilepsy.
- In the treatment of primary generalized tonic-clonic seizures in adults and children from 12 years with Idiopathic Generalized Epilepsy.

Recommended dosage

Route of administration

Oral

Monotherapy in adults and adolescents over 16 years

The recommended starting dose is 250 mg twice daily, which should be increased to an initial therapeutic dose of 500 mg twice daily after two weeks of treatment. The dose may be increased according to clinical response with 250 mg twice daily every 2 weeks. The maximum dose is 1,500 mg twice daily.

Concomitant treatment in adults (>18 years) and adolescents (12-17 years) weighing 50 kg or more

The initial therapeutic dose is 500 mg twice daily. This dose can be started from the first day of treatment. Depending on the clinical response and tolerability, the daily dose can be increased to 1500 mg twice daily. The dose modification can be performed with increases or reductions of 500 mg twice daily every two to four weeks.

Special Population Groups

Elderly (65 years and older)

It is recommended to adjust the dose in elderly patients with impaired kidney (renal) function (see "renal impairment").

Renal impairment

The daily dose should be individualized according to renal function. The following table indicates how to adjust the dose in adult patients. In order to use this dosage table, an estimate of creatinine ratio (CL_{cr}) is needed, in ml/mm, of the patient. The CL_{cr} in ml/mm may be estimated from the measurement of serum creatinine (mg/dl) for adults and adolescents weighing 50 kg or more by using the formula:

$$CL_{cr} \text{ (ml/min)} = \frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dl)}} \times 0.85 \text{ for women}$$

CL_{cr} is then fitted to the body surface area (BSA) as follows:

$$CL_{cr} \text{ (ml/min/1.73 m}^2\text{)} = \frac{CL_{cr} \text{ (ml/min)}}{\text{Person's BSA (m}^2\text{)}} \times 1.73$$

Dosing adjustment for adult and adolescent patients weighing more than 50 kg with renal impairment

Group	Creatinine clearance (ml/min/1.73m ²)	Dose and frequency
Normal	> 80	500 to 1,500 mg twice daily
Mild	50-79	500 to 1,000 mg twice daily
Moderate	30-49	250 to 750 mg twice daily
Severe	< 30	250 to 500 mg twice daily
Patients with disease ESRD on dialysis ⁽¹⁾	-	500 to 1,000 mg once daily

⁽¹⁾ A loading dose of 750 mg is recommended in the first day of treatment with levetiracetam.

⁽²⁾ After dialysis, a supplemental dose of 250 to 500 mg is recommended.

In children with renal impairment, the levetiracetam dose needs to be adjusted based on renal function as levetiracetam clearance is related to renal function. This recommendation is based on a study in adult patients with renal impairment.

The CL_{cr} in ml/min/1.73 m² can be estimated from the measurement of serum creatinine (mg/dl) for young adolescents, children and breast-feeding infants, using the following formula (Schwartz formula):

$$CL_{cr} \text{ (ml/min/1.73 m}^2\text{)} = \frac{\text{Height (cm)} \times \text{ks}}{\text{Serum creatinine (mg/dl)}}$$

ks = 0.55 in children under 13 and adolescent female; ks = 0.7 in adolescent male. Dosing adjustment for infants, children and adolescents weighing less than 50kg with impaired renal function:

Group	Creatinine clearance (ml/min/1.73m ²)	Children and adolescents weighing less than 50 kg
Normal	> 80	10 to 30 mg/kg (0.10 to 0.30 ml/kg) twice daily
Mild	50-79	10 to 20 mg/kg (0.10 to 0.20 ml/kg) twice daily
Moderate	30-49	5 to 15 mg/kg (0.05 to 0.15 ml/kg) twice daily
Severe	< 30	5 to 10 mg/kg (0.05 to 0.10 ml/kg) twice daily
End-stage renal disease patients undergoing dialysis	-	10 to 20 mg/kg (0.10 to 0.20 ml/kg) once daily ⁽²⁾⁽³⁾

⁽¹⁾ levetiracetam oral solution should be used for doses below 250 mg and for patients who cannot swallow tablets.

⁽²⁾ A 15 mg/kg (0.15 ml/kg) loading dose is recommended in the first day of treatment with levetiracetam.

⁽³⁾ Following dialysis, a supplemental dose of 5 to 10 mg/kg (0.05 to 0.10 ml/kg) is recommended.

Hepatic impairment

No need to adjust dose in patients with mild to moderate hepatic impairment. In patients with severe hepatic impairment, the creatinine clearance may underestimate the degree of renal insufficiency. Therefore, a reduction of 50% of the daily maintenance dose is recommended when the creatinine clearance is <60 ml/min/1.73 m².

Paediatric population

The physician should prescribe the pharmaceutical form, presentation and most appropriate concentration according to age, weight and dose. The tablet formulation is not adapted for use in breast-feeding infants and children less than 6 years of age. Levetiracetam oral solution is the most appropriate formulation for administration in this population. In addition, the concentrations of available tablets are not appropriate for initial treatment in children weighing less than 25 kg, for patients unable to swallow tablets or for the administration of doses below 250 mg. In all these cases levetiracetam oral solution should be used. Levetiracetam oral solution is the preferred formulation for use in children under the age of 6 years.

For children 6 years and above, levetiracetam oral solution should be used for doses under 250 mg, for doses not multiple of 250 mg when dosing recommendation is not achievable by taking multiple tablets and for patients unable to swallow tablets.

Monotherapy

The safety and effectiveness of levetivae as monotherapy in children and adolescents under 16 years has not been established. No data available.

Add-on therapy for children (4 to 11 years) and adolescents (12 to 17 years) weighing less than 50 kg

Levetiracetam oral solution is the preferred formulation for use in infants and children under the age of 6 years. The safety and efficacy of levetiracetam concentrate for solution for infusion in infants and children less than 4 years have not been established.

The initial therapeutic dose is 10 mg/kg twice daily.

Depending upon the clinical response and tolerability, the dose can be increased up to 30mg/kg twice daily. Dose changes should not exceed increases or decreases of 10 mg/kg twice daily every two weeks. The lowest effective dose should be used. Dose in children 50 kg or greater is the same as in adults.

Dose recommendations for children and adolescents:

Weight	Starting dose 10 mg/kg twice daily	Maximum dose 30 mg/kg twice daily
10 kg ⁽¹⁾	100 mg (1 ml) twice daily	300 mg (3 ml) twice daily
15 kg ⁽¹⁾	150 mg (1.5 ml) twice daily	450 mg (4.5 ml) twice daily
20 kg ⁽¹⁾	200 mg (2 ml) twice daily	600 mg (6 ml) twice daily
25 kg	250 mg twice daily	750 mg twice daily
From 50 kg ⁽²⁾	500 mg twice daily	1500 mg twice daily

⁽¹⁾ Children 25 kg or less should preferably start the treatment with levetiracetam 100 mg/ml oral solution

⁽²⁾ Dose in children and adolescents 50 kg or more is the same as in adults

Form of administration

The film-coated tablets are administered orally, with a sufficient quantity of liquid and may be taken with or without food. The daily dose is divided in equal doses divided into two doses daily. The consumer should swallow the tablet whole and not divide the tablet to obtain equal doses.

Contraindications

Hypersensitivity to the active substance, to other pyrrolidone derivatives or any of the excipients.

Warning and precautions

Discontinuation of treatment

According to routine clinical practice, if the Levetiracetam medication needs to be stopped, it is recommended to withdraw it gradually (e.g., in adults and adolescents weighing more than 50 kg: Reductions of 500 mg twice daily every two to four weeks, in breast-feeding infants older than 6 months, children and adolescents weighing less than 50 kg: the dose decrease should not exceed 10 mg/kg twice daily every two weeks; in breast-feeding infants under 6 months: dose decrease should not exceed 7 mg/kg twice daily every two weeks).

Kidney failure

The levetitae administration to patients with kidney failure may require dose adjustment. In patients with severe liver failure renal function should be evaluated prior to dose selection.

Acute kidney failure

The use of levetiracetam has been rarely associated with acute kidney injury, with a time to onset ranging from a few days to several months.

Blood cell counts

Rare cases of decreased blood cell counts (neutropenia, agranulocytosis, leucopenia, thrombocytopenia and pancytopenia) have been described in association with levetiracetam administration, generally at the beginning of the treatment. Complete blood cell counts are advised in patients experiencing important weakness, pyrexia, recurrent infections or coagulation disorders

Suicide

Potential for an increase in risk of suicidal thoughts or behaviors. Suicide cases have been reported, suicide attempt and suicidal thoughts and behaviour in patients treated with anti-epileptic medication (including levetiracetam). A meta-analysis of placebo-controlled trials, randomized, with anti-epileptic drugs has shown a small increased risk of suicidal thoughts and behaviour. The mechanism of this risk is unknown.

Therefore, patients should be monitored for signs of depression and/or suicidal thoughts and behaviours and appropriate treatment should be considered. Patients (and their caretakers) should be advice to contact their doctor if signs of depression and/or suicidal thoughts appear.

Paediatric population

The tablet formulation is not adapted for use in breast-feeding infants and children less than 6 years of age.

Available data in children did not suggest any impact on growth and puberty. However, the long-term effects on learning, intelligence, growth, endocrine function, puberty and childbearing potential in children remain unknown.

Interaction with other medicaments

Anti-epileptic medication Data from clinical studies conducted in adults before marketing indicate that levetiracetam does not influence the serum concentrations of existing anti-epileptic medicinal products (phenytoin, carbamazepine, valproic acid, phenobarbital, lamotrigine, gabapentin and primidone) and that these anti-epileptic drugs not influence the pharmacokinetics of levetiracetam. As in adults, there is no clear evidence of clinically significant drug interactions in paediatric patients receiving up to 60mg/kg/day levetiracetam. A retrospective assessment of pharmacokinetics interactions in children and adolescents with epilepsy (4 to 17 years) confirmed that coadjunctive treatment with orally administered levetiracetam did not influence the serum concentrations at steady state of carbamazepine and valproate. However, the datasuggest an increased 20% clearance of levetiracetam in children taking anti-epileptic medicinal enzyme inducers. Dose adjustment is not needed.

Probenecid

It has been found that probenecid (500 mg four times daily), a blocking agent of renal tubular secretion, inhibits the renal clearance of the primary metabolite, but not the levetiracetam one. However, the levels of this metabolite remain low.

Methotrexate

Concomitant administration of levetiracetam and methotrexate has been reported to decrease methotrexate clearance, resulting in increased/prolonged blood methotrexate concentration to potentially toxic levels. Blood methotrexate and levetiracetam levels should be carefully monitored in patients treated concomitantly with the two drugs.

Laxatives

There have been isolated reports of decreased levetiracetam efficacy when the osmotic laxative macrogol has been concomitantly administered with oral levetiracetam. Therefore, macrogol should not be taken orally for one hour before and for one hour after taking levetiracetam.

Oral contraceptives and other pharmacokinetic interactions

Daily doses of 1000 mg levetiracetam did not influence the pharmacokinetics of oral contraceptives (ethinyl estradiol and levonorgestrel); endocrine parameters (luteinizing hormone and progesterone) were not modified. Daily doses of 2,000 mg levetiracetam did not influence the pharmacokinetics of digoxin and warfarin; prothrombin time remained unchanged. Co-administration with digoxin, oral contraceptives and warfarin did not influence the pharmacokinetics of levetiracetam.

Food and alcohol

The extent of absorption of levetiracetam did not change with food intake, although the rate of absorption was slightly reduced. There is no data on the interaction of levetiracetam with alcohol.

Statement on usage during pregnancy and lactation

Pregnancy

Postmarketing data from several prospective pregnancy registries have documented outcomes in over 1,000 women exposed to levetiracetam monotherapy during the first trimester of pregnancy. Overall, these data do not suggest a substantial increase in the risk for major congenital malformations, although a teratogenic risk cannot be completely excluded. Therapy with multiple antiepileptic medicinal products is associated with a higher risk of congenital malformations than monotherapy and, therefore, monotherapy should be considered. Studies in animals have shown reproductive toxicity. Levetiracetam is not recommended during pregnancy and in women of childbearing potential not using contraception unless clinically necessary.

Physiological changes during pregnancy may affect levetiracetam concentration. Decrease in levetiracetam plasma concentrations has been observed during pregnancy. This decrease is more pronounced during the third trimester (up to 60% of baseline concentration before pregnancy). Appropriate clinical management of pregnant women treated with levetiracetam should be ensured. Discontinuation of antiepileptic treatments may result in exacerbation of the disease which could be harmful to the mother and the foetus.

Breastfeeding

Levetiracetam is excreted in human breast milk, so breastfeeding is not recommended. However, if levetiracetam treatment is desired during breastfeeding, consider the benefit/risk ratio of treatment taking into account the importance of breastfeeding.

Fertility

In animal studies no impact on fertility was detected. No clinical data are available; the potential risk for humans is unknown.

Adverse effects/Undesirable effects

Summary of the safety profile

The most frequently reported adverse reactions were nasopharyngitis, somnolence, headache, fatigue and dizziness. The adverse reaction profile presented below is based on the analysis of pooled

placebo-controlled clinical trials with all indications studied, with a total of 3,416 patients treated with levetiracetam. These data are supplemented with the use of levetiracetam in corresponding open-label extension studies, as well as post-marketing experience. The safety profile of levetiracetam is generally similar across age groups (adult and paediatric patients) and across the approved epilepsy indications.

Tabulated list of adverse reactions

MedDRA SOC	Frequency category			
	Very common	Common	Uncommon	Rare
Infections and infestations	Nasopharyngitis			Infection
Blood and lymphatic system disorders			Thrombocytopenia, leukopenia	Pancytopenia, neutropenia, agranulocytosis
Immune system disorders				Drug reaction with eosinophilia and systemic symptoms (DRESS), Hypersensitivity (including angioedema and anaphylaxis)
Metabolism and nutrition disorders		Anorexia	Weight decreased, weight increase	Hyponatraemia
Psychiatric disorders		Depression, hostility/ aggression, anxiety, insomnia, nervousness/ irritability	Suicide attempt, suicidal ideation, psychotic disorder, abnormal behaviour, hallucination anger, confusional state, panic attack, affect liability/mood swings, agitation	Completed suicide, personality disorder, thinking abnormal
Nervous system disorders	Somnolence, headache	Convulsion, balance disorder, dizziness, lethargy, tremor	Amnesia, memory impairment, coordination abnormal/ataxia, paraesthesia, disturbance in attention	Choreoathetosis, dyskinesia, hyperkinesia
Eye disorders			Diplopia, vision blurred	
Ear and labyrinth disorders		Vertigo		
Respiratory, thoracic and mediastinal disorders		Cough		
Gastrointestinal disorders		Abdominal pain, diarrhoea, dyspepsia, vomiting, nausea		Pancreatitis
Hepatobiliary disorders			Liver function test abnormal	Hepatic failure, hepatitis
Renal and Urinary disorders				Acute Kidney injury
Skin and subcutaneous tissue disorders		Rash	Alopecia, eczema, pruritus	Toxic epidermal necrolysis, Stevens-Johnson syndrome, erythema multiforme
Musculoskeletal and connective tissue disorders			Muscular weakness, myalgia	Rhabdomyolysis and blood creatine phosphokinase increased*
General disorders and administration site conditions		Asthenia/ fatigue		
Injury, poisoning and procedural complications			Injury	

*Prevalence is significantly higher in Japanese patients when compared to non-Japanese patients. Cases of encephalopathy have been rarely observed after levetiracetam administration. These undesirable effects generally occurred at the beginning of the treatment (few days to a few months) and were reversible after treatment discontinuation"

Description of some selected adverse reactions

The risk of anorexia is higher when topiramate is co-administered with levetiracetam.

In several cases of alopecia, recovery after discontinuation of treatment with levetiracetam was observed.

Bone marrow suppression was identified in some of the cases of pancytopenia.

Paediatric population

In patients aged 1 month to less than 4 years, a total of 190 patients have been treated with levetiracetam in placebocontrolled and open label extension studies. Sixty of these patients were treated with levetiracetam in placebo-controlled studies. In patients aged 4-16 years, a total of 645 patients have been treated with levetiracetam in placebo-controlled and open label extension studies. 233 of these patients were treated with levetiracetam in placebo-controlled studies. In both these paediatric age ranges, these data are supplemented with the post-marketing experience of the use of levetiracetam.

In addition, 101 infants aged less than 12 months have been exposed in a post authorization safety study. No new safety concerns for levetiracetam were identified for infants less than 12 months of age with epilepsy.

The adverse reaction profile of levetiracetam is generally similar across age groups and across the approved epilepsy indications. Safety results in paediatric patients in placebo-controlled clinical studies were consistent with the safety profile of levetiracetam in adults except for behavioural and psychiatric adverse reactions which were more common in children than in adults. Results related to behavioural and emotional functioning indicated a worsening in levetiracetamtreated patients on aggressive behaviour as measured in a standardised and systematic way using a validated instrument (CBCL – Achenbach Child Behavior Checklist). However subjects, who took levetiracetam in the long-term open label follow-up study, did not experience a worsening, on average, in their behavioural and emotional functioning; in particular measures of aggressive behaviour were not worse than baseline.

Overdose and treatment

Symptoms

The following have been observed: drowsiness, agitation, aggressiveness, decreased level of consciousness, respiratory depression and coma with levetitae overdoses.

Management of overdose

In acute overdose the stomach may be emptied by gastric lavage or by emesis induction (vomiting). There is no specific antidote for levetiracetam. Treatment of an overdose will be symptomatic and may include haemodialysis. The efficiency of removal by dialysis is 60% for levetiracetam and 74% for the primary metabolite.

Effects in Ability to Drive and Use Machine

Levetiracetam has minor or moderate influence on the ability to drive and use machines. Due to possible different individual sensitivity, some patients might experience somnolence or other central nervous system related symptoms, especially at the beginning of treatment or following a dose increase. Therefore, caution is recommended in those patients when performing skilled tasks, e.g. driving vehicles or operating machinery. Patients are advised not to drive or use machines until it is established that their ability to perform such activities is not affected.

Storage conditions

Store at room temperature below 30°C. **Dosage forms and packaging available.**

levevitae 500 mg is packaged in aluminium/PVC/PVDC blister and is available in 30 and 100 film-coated tablet packs.

Manufacturer:

Pharmathen S.A. – Dervenakion 6, Pallini 15351, Attikis, Greece

Product registration holder

Healol Pharmaceuticals Sdn. Bhd.
74-3, Jalan Wangsa Delima 6,
KLSC Wangsa Maju, 53300 Kuala Lumpur, Malaysia

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10 November 2020

MOCK UP

Galenicum believe in life

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